

MATH 126 ESSENTIALS

1. CHAIN RULE

Suppose that $F : \mathbb{R}^n \rightarrow \mathbb{R}^k$,

$$F(x) = (F_1(x), F_2(x), \dots, F_k(x)), \quad F_j : \mathbb{R}^n \rightarrow \mathbb{R},$$

and $G : \mathbb{R}^k \rightarrow \mathbb{R}^m$,

$$G(y) = (G_1(y), \dots, G_m(y)), \quad G_j : \mathbb{R}^k \rightarrow \mathbb{R}^m.$$

We can then obtain a function

$$H : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad H(x) = G(F(x)), \quad H(x) = (H_1(x), \dots, H_m(x)), \quad H_j(x) = G_j(F(x)).$$

Then for $1 \leq j \leq m$, and $1 \leq \ell \leq n$,

$$\frac{\partial H_j(x)}{\partial x_\ell}(x) = \sum_{p=1}^k \frac{\partial G_j(F(x))}{\partial y_p} \frac{\partial F_p(x)}{\partial x_\ell}.$$

If we wrote $y = F(x)$ then a *shorthand* for this is

$$\frac{\partial}{\partial x_\ell} = \sum_{p=1}^k \frac{\partial y_p}{\partial x_\ell} \frac{\partial}{\partial y_p}, \quad y_p = F_p(x).$$

2. FIRST ORDER EQUATIONS

2.1. Linear and semilinear equations. Method of characteristics for semilinear equations is the following algorithm: We solve

$$\begin{aligned} \partial_t u + v(t, x) \partial_x u + w(t, x, u(t, x)) &= 0 \\ u(0, x) &= f(x) \end{aligned}$$

- (1) Solve $x'(t) = v(t, x(t))$ for characteristic $x(t)$ with initial value $x(0)$
- (2) Solve $h'(t) + w(t, x(t), h(t)) = 0$ for $h(t)$, with initial value $h(t) = f(x(0))$.
- (3) Match $u(t, x(t)) = h(t)$ and $h(0) = f(x(0))$ where $x(t) = x$.

The equation is linear when w does *not* depend on u . A similar approach works in higher dimensions, $x \in \mathbb{R}^n$, in which case $v(t, x)$ is a vector valued functions and we solve a system of ODE (see p.29 in the notes)

2.2. Quasilinear equations.

$$\begin{aligned}\partial_t u + a(u) \cdot \nabla u &= 0 \\ u(0, x) &= f(x)\end{aligned}$$

Here is the procedure for finding solutions:

- (1) Solution is given *implicitly* by $u(t, y + ta(f(y))) = f(y)$.
- (2) For a given t and x we want to find y such that $x = y + ta(f(y))$, $y = y(t, x)$.
- (3) If we can find such $y = y(t, x)$ then $u(t, x) = f(y(t, x))$.

The difficulty is in the second step: we know that for small t a solution exists (from the implicit function theorem – see p.34) but even then it might be hard to write down.

3. WAVE EQUATION

You need to know

- Expression for solutions of the initial value problem when $x \in \mathbb{R}^n$, $n = 1, 2, 3$ (pp. 40, 52, 55); except when $n = 1$ you should not worry about the derivation of these formulas.
- Duhamel's formula (p.46) (preferably understanding how it can be applied in higher dimensions as well).
- Boundary value problems in dimension one (p.44)
- Huygens principle (pp.54, 57-58); there will be no tricky questions about it but it is something you should try to remember.

4. SEPARATION OF VARIABLES

Here you should know the following things:

- General principle behind separation of variables (p.63); its relevance to solving the heat (and wave) equation using Fourier series (p.95)
- Separation of variables in rectangular coordinates
- Separation of variables in polar coordinates; eigenvalues of the Laplacian for the disc, Bessel functions (pp 66–67)
- Spherical harmonics: no need for memorizing formulas and derivations but you should know the parametrization of spherical harmonics (p.71)
- Solutions to the wave (and heat) equation using *eigenvalues* of the Laplacian (p.68, p.75)

You should also understand (see p.65) why (for Ω bounded with a piecewise C^1 boundary)

$$-\Delta\varphi = \lambda\varphi, \quad \text{in } \Omega, \quad \varphi|_{\partial\Omega} = 0, \quad \varphi \not\equiv 0 \implies \lambda > 0,$$

and

$$-\Delta\varphi = \lambda\varphi, \quad \text{in } \Omega, \quad \partial_\nu\varphi|_{\partial\Omega} = 0, \quad \varphi \not\equiv 0 \implies \lambda \geq 0.$$

5. HEAT EQUATION

You should remember the *heat kernel* in any dimension

$$H_t(x) := (4\pi t)^{-\frac{n}{2}} \exp(-|x|^2/4t), \quad t > 0, \quad x \in \mathbb{R}^n$$

and the role it plays in solving the initial value problem (p.82) and the non-homogeneous problem (p.81) (there is no need to memorize the lengthy derivation but it would be nice if you understood it). It is important to note that the principle works in all dimensions.

6. FUNCTION SPACES

Here you should focus on Hilbert spaces and Fourier series.

6.1. **Hilbert spaces.** You need to know what are

- inner products and Cauchy Schwartz inequality, parallelogram identity
- orthonormal sets
- Bessel inequality, including the condition for the equality (p.92)
- orthonormal basis: definition and characterization on p.93

6.2. **Fourier series.** You need to know and be able to prove that

$$\{L^{-\frac{1}{2}}e^{2\pi i kx/L}\}_{k \in \mathbb{Z}}, \quad \{(2/L)^{\frac{1}{2}}\sin(\pi kx/L)\}_{k=1}^\infty, \quad \{L^{-\frac{1}{2}}\} \cup \{(2/L)^{\frac{1}{2}}\cos(\pi kx/L)\}_{k=1}^\infty,$$

are orthonormal sets in $L^2([0, L])$.

You need to know (but do not have to prove!) that these sets are also orthonormal basis.

You need to know how to apply abstract results (such as the Bessel inequality or expansion in terms of a basis) in the context of $L^2([0, L])$

And, finally you need to know how to use Fourier series and separation of variables to solve PDE (p.95).